Business Challenge Cohort Analysis

Anirudh Unni, Raissa Franco & Ayse Kocyigit

Project Overview

- **Goal:** Understand the long-term behavior of IronHack Payments users by grouping them into cohorts based on their acquisition month.

  Key Objectives:

- Track how user engagement changes over time
- Identify which user groups are the most valuable
- Measure key performance metrics like usage frequency, payment incident rates, and revenue per cohort

EDA & QA

- Taking a critical look at the given data
- Created a unified user ID and removed irrelevant columns
- Cleaned missing values (e.g., user_id, send_at)
- Converted timestamps to datetime and set monthly periods
- Combined two datasets: cash requests + fees

Dataframe cash_request

```
# It converts the 'send_at' column in the DataFrame cash_df into datetime format
cash_df['send_at'] = pd.to_datetime(cash_df['send_at'], format='mixed')
```

```
# For creating the new column 'cohort':
# for each user group, we take the minimum 'send_at' date
# we convert it to a monthly period
cash_df['cohort'] = cash_df.groupby('all_user_id')['send_at'].transform('min').dt.to_period('M')
```

```
cash_df.head()
```

	id	amount	status	deleted_account_id	send_at	recovery_status	all_user_id
48	23534	25.0	money_back	False	2020-10-23 15:21:26.878525+00:00	No incident	21465.0
55	9323	50.0	rejected	False	2020-07-28 21:33:12.430087+00:00	No incident	23318.0
60	18701	100.0	rejected	False	2020-10-08 11:41:07.170136+00:00	No incident	57285.0
62	16121	50.0	rejected	False	2020-09-21 23:05:32.558375+00:00	No incident	38075.0
135	18459	100.0	rejected	False	2020-10-06 18:36:25.849957+00:00	No incident	55584.0

Dataframe data_fees

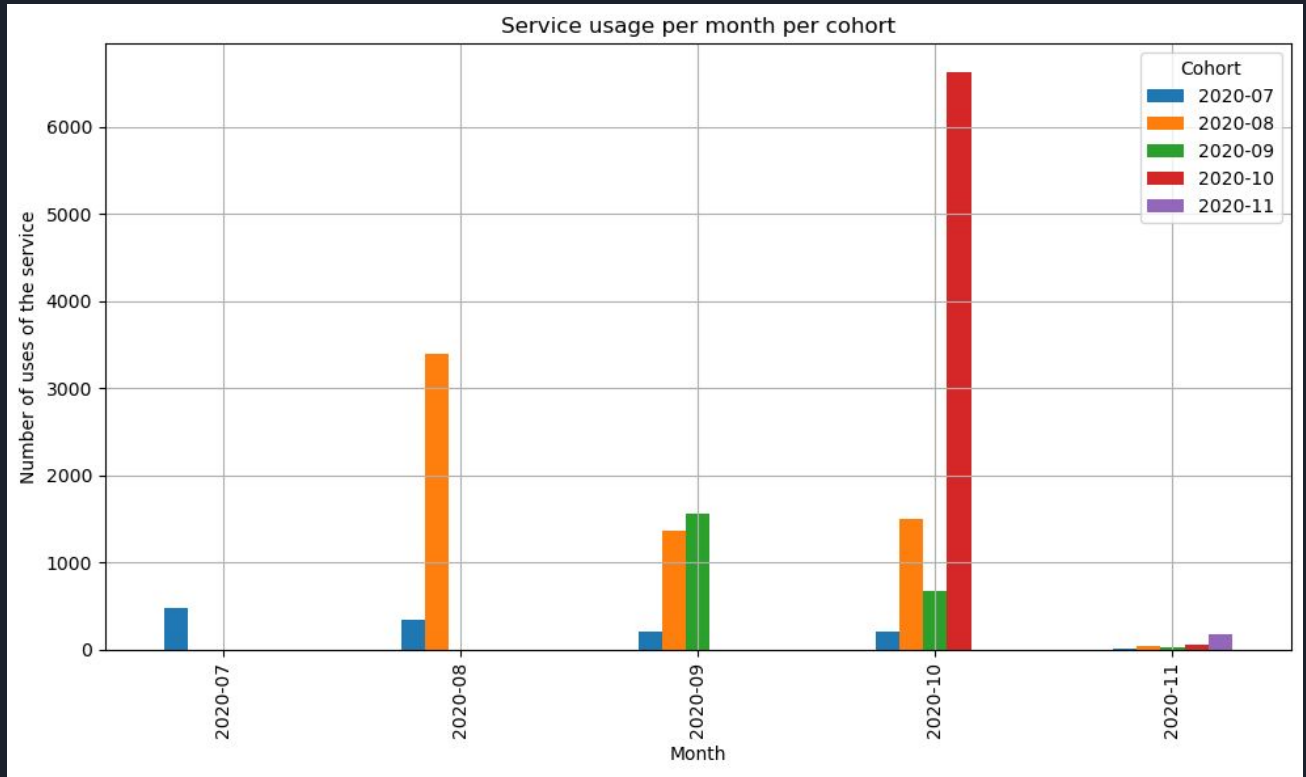
(21057, 7)

	id	cash_request_id	type	status	category	total_amount	paid_at
0	6537	14941.0	instant_payment	rejected	NaN	5.0	2020-12-17 14:50:07.47011+00
1	6961	11714.0	incident	accepted	rejected_direct_debit	5.0	2020-12-08 17:13:10.45908+00
2	16296	23371.0	instant_payment	accepted	NaN	5.0	2020-11-04 19:34:37.43291+00
3	20775	26772.0	instant_payment	accepted	NaN	5.0	2020-11-19 05:09:22.500223+00
4	11242	19350.0	instant_payment	accepted	NaN	5.0	2020-11-02 14:45:20.355598+00

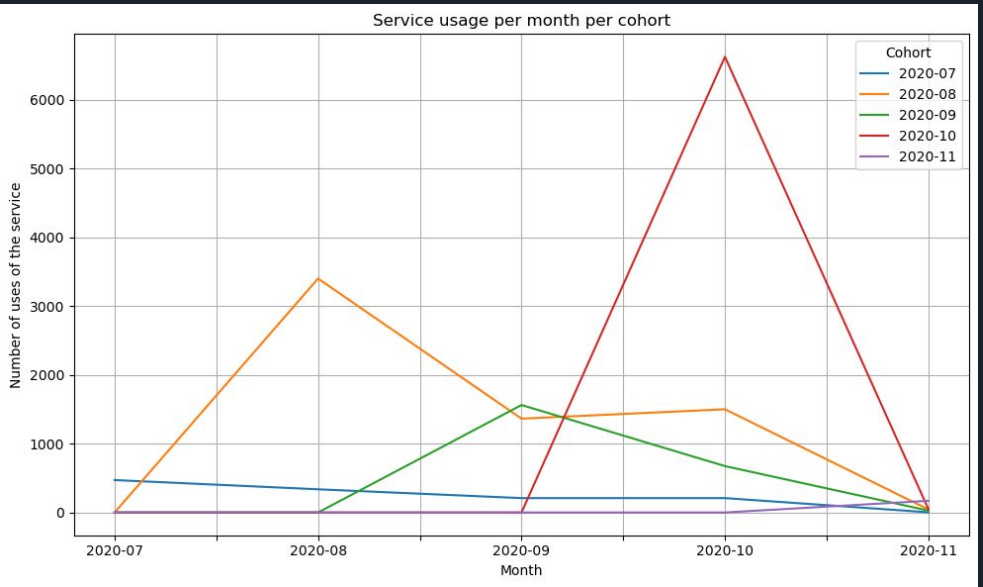
1st metric: Frequency of service usage

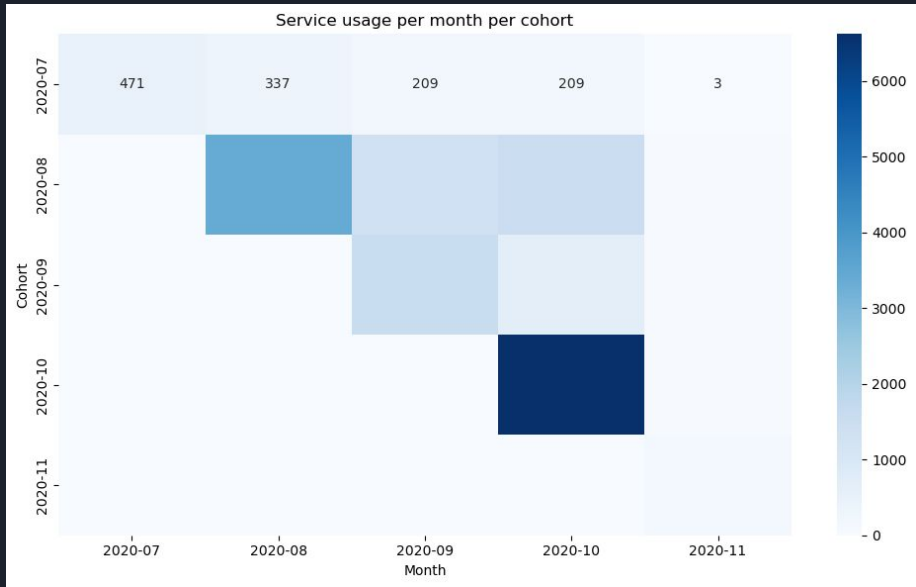
Insights:

- A drop in the number of users within each cohort over time
- Most users start using the cohort in October
- Significant drop in users in November



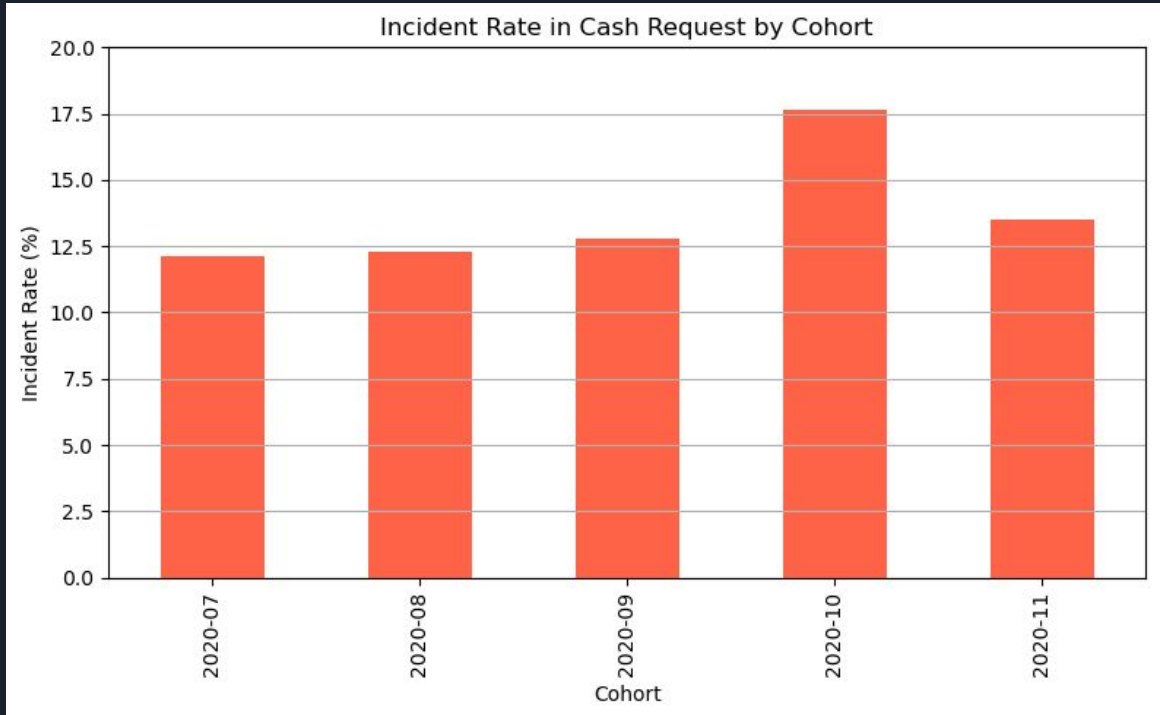
Different types of visualizations





2nd metric: Incident rate

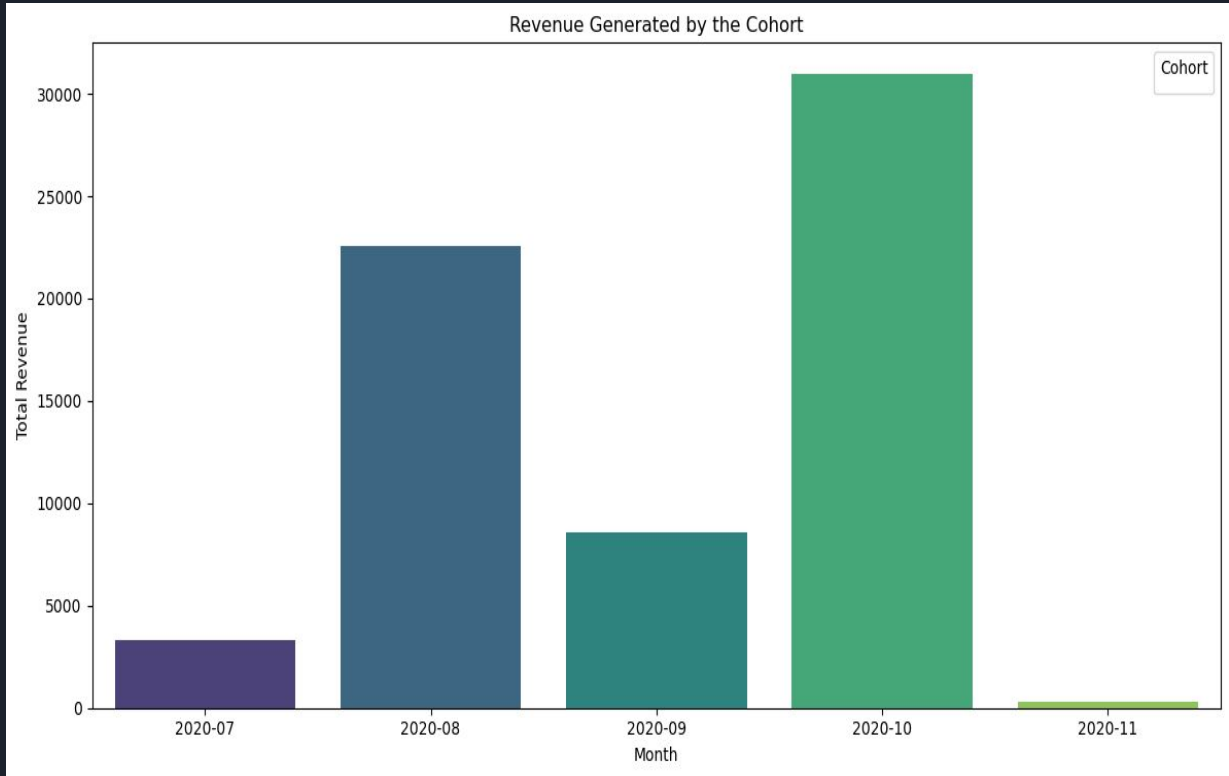
- Insights:
  - Higher number of incident rates in October compared to other cohorts.
- Interpretation:
  - The number of users in October were significantly higher than other cohorts, resulting in higher incident rates in CR.



3rd metric: Revenue generated by the cohort

Insights

- Revenue directly proportional to the no. of users in cohort.
- Maximum revenue in October, least in November.



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